

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M.Tech. (EC)(C.S.E.) Sem. - I Examination, 2010-11
EC- 705 Advanced Digital Satellite Communications

Date: 07/01/2011, Friday
Maximum Marks: 70

Time: 10:30 am to 1:30 pm

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION- I

- Q-1 a) State kepler's three laws of planetary motion. 3
- b) Define the following terms: 4
- Critical frequency, prograde orbit, Polarization interleaving, cross polarization discrimination
- c) Explain how depolarization is caused by rain. 2
- d) A satellite is operated at EIRP of 56 db with an output back off of 6 db. The transmitter feeder losses to 2 db and the antenna gain is 50 db. Calculate the power output of the TWTA, assuming it may be require providing the full saturated EIRP. 2
- Q-2 a) Define *effective isotropic radiated power*. A transmitter feeds a power of 10 W into an antenna which has a gain of 46 db. Calculate the EIRP in i) Watts ii) dBW. 3
- b) A satellite is orbiting in the equatorial plane with a period from perigee to perigee of 12hr. Given that the eccentricity is 0.002. Calculate the semi major axis. The earth equatorial radius is 6378.1414km. ($\mu=3.986005 \times 10^{14} \text{ m}^3/\text{cm}^2$) 3
- c) Describe and compare the master antenna TV system and community antenna TV system. 4
- d) Find the time in Julian Centuries from the reference time January 0.5, 1900 to 2

13h UT on 18 December 2000.

OR

- Q-2 a) Define *saturation flux density* and prove that 6
- $$[EIRP_s]_U = [\Psi_s] + [A_0] + [LOSSES]_U - [RFL]$$
- b) Describe telemetry, tracking, and command facilities of a satellite communication system. 6
- Q-3 a) Explain how intermodulation noise originates in a satellite link, and describe how it can be reduced. In a satellite circuit the carrier to noise spectral density ratios are: uplink 23dBHz, downlink 20 dBHz, intermodulation 24dBHz. Calculate the overall carrier to noise ratio. 5
- b) A circularly polarized wave at frequency of 12GHz is transmitted from satellite. The point rain rate is 13mm/h. calculate the specific attenuation. ($a_h=0.0188$, $a_v=0.168$, $b_h=1.217$, $b_v=1.2$ for $f=12\text{GHz}$) 2
- c) Explain various causes and effects of transmission losses occurring in satellite communication system. 5

OR

- Q-3 a) Determine the azimuth angle for a geostationary satellite at 90 degrees W, for an earth station at latitude 35 degree N, longitude 100 degrees W. 4
- b) Compare the uplink power requirement for Time Division Multiple Access and Frequency Division Multiple Access techniques. 3
- c) Given that $L_U=200\text{db}$, $L_D=196\text{db}$, $G_E=G'_E=25\text{db}$, $G_S=G'_S=9\text{db}$, $G_{TE}=G_{RE}=48\text{db}$, $G_{RS}=G_{TS}=19\text{db}$, $U_S=U'_S=1\mu\text{J}$, and $U_E=U'_E=10\mu\text{J}$, calculate the transmission gain $[\gamma]$, interference level $[I_1]$ and $[I_2]$, and the equivalent temperature rise overall. 5

SECTION- II

- Q-4 a) Write technical note on time division multiplexing and discuss various errors associated with it. 4
- b) What do you mean by *energy dispersal*? 2
- c) Explain why the low noise amplifier in a satellite receiving system is placed at the antenna end of feeder cable. 3

- d) A satellite link operating at 14 GHz has receiver feeder losses of 1.5db and a free space loss of 207 db. The atmospheric absorption loss is 0.5db and the antenna pointing loss is 0.5db. Depolarization losses may be neglected. Calculate the total link loss for clear sky conditions. 2

- Q-5 a) Briefly discuss the equipment sections making up a transponder channel. 5
- b) Describe the channeling scheme for SPADE system for Intelsat satellite. Discuss its DASS based communication system. 5
- c) Determine the orbit eccentricity of a satellite moving in elliptical path having semi major axis equal to 16000 km if the difference between perigee and apogee is 25000 km and earth radius is 6360 km. 2

OR

- Q-5 a) Explain what is meant by satellite attitude and brief describe two forms of attitude control. 5
- b) Describe the principle of operation of Code Division Multiple Access. What effects do the unwanted signals have on the wanted signal? 4
- c) Explain why a minimum of four satellites must be visible at any earth location utilizing the GPS system for position determination. 3

Q-6 Attempt any three 12

- a) Describe main features and services offered by the IRIDIUM satellite system.
- b) Describe the operation of typical VSAT system.
- c) Describe the main features of RADARSAT. Explain what is meant by a 'down to dusk' orbit and why the RADARSAT follows such orbit.
- d) Discuss briefly the need for MSAT system, and explain how this compares with existing terrestrial cellular networks.
- e) Draw and explain Receive-Only Home TV systems.